Boundary K-matrices at every Fuss–Catalan rank

Zhiyuan Yao[1, *]

[1] *Lanzhou Center for Theoretical Physics, Key Laboratory of Theoretical Physics of Gansu Province, Key Laboratory of Quantum Theory and Applications of MoE, Gansu Provincial Research Center for Basic Disciplines of Quantum Physics, Lanzhou University, Lanzhou 730000, China*

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The Fuss–Catalan algebras replace each site of a Temperley–Lieb chain by a bundle of r nested coloured strands, and Di Francesco's Baxterization turns them into integrable lattice models of dense coloured loops for every r. Integrability in a strip needs a boundary operator obeying the reflection equation as well, and such an operator was known only for one and two colours. We classify, for every integer $r \geq 3$ and generic loop weights, the identity-normalized meromorphic operators $K(w) = 1 + \sum_{s=1}^{r} k_s(w)E^{(s)}$ that satisfy the reflection equation together with $K(w)K(w^{-1}) = 1$, where $E^{(s)}$ attaches the innermost s strands of the boundary bundle to the wall. Every solution has $k_s \equiv 0$ for $s \geq 3$, so no boundary operator ever reaches past the second colour, however many colours the model carries. Over the field $\mathbb{C}(q, q_2)$ generated by the two loop-weight parameters the complete list is the identity together with a one-constant family supported on $E^{(1)}$; adjoining one square root, which is a genuine quadratic extension, adds exactly two further conjugate branches supported on $E^{(1)}$ and $E^{(2)}$. The list is the same at every rank. The proof combines an exhaustive 36-diagram elimination at rank three with an injective colour-tail homomorphism that removes one colour at a time, and a four-colour obstruction that, transported by a pure-boundary projection, forbids the top colour at every rank. The result fixes the boundary conditions available to double-row transfer matrices built on these models, and shows that the multi-colour boundary problem is far more rigid than its single-colour counterpart.

I. INTRODUCTION

A large part of two-dimensional solvable lattice physics is organized by one algebra. The Temperley–Lieb algebra [1] is generated by elements $e_1, \ldots, e_{n-1}$, one for each pair of neighbouring sites, and its content is diagrammatic: an element is a planar pairing of n points on a bottom edge with n points on a top edge by non-crossing strands, multiplication is vertical stacking, and each closed loop the stacking creates is erased in exchange for a scalar loop weight [2]. Replacing a generator by a spectral-parameter-dependent combination of itself and the identity — a procedure known as Baxterization — turns any such algebra into a solution of the Yang–Baxter equation, hence into a family of commuting transfer matrices and an exactly solvable model [3, 4].

Bisch and Jones [5] found a rigid multi-colour generalization of that algebra while studying intermediate subfactors. In the resulting Fuss–Catalan algebras each site carries a bundle of r nested coloured strands, and the generator labelled s, for $1 \leq s \leq r$, reconnects the innermost s strands of two neighbouring bundles while the outer $r - s$ strands run straight through; one colour returns the Temperley–Lieb algebra. Landau described these algebras completely, giving a basis in terms of generators and the dimensions of the irreducible representations, and showed that they sit inside the higher relative commutants of any subfactor containing a chain of intermediate subfactors [6]; their representation theory has been developed since [7], as has their relation to quantum groups [8]. Di Francesco [9] showed that these algebras Baxterize as well, and classified the resulting bulk Yang–Baxter solutions for every r: for each rank there is one generic branch, which he called fundamental, alongside degenerate ones that first appear at four colours. The lattice models are gases of dense coloured loops. The algebraic approach to solvable lattice models has since been pursued more broadly [10], and Poncini and Rasmussen determined which singly generated planar algebras underlie a Yang–Baxter integrable model with a homogeneous transfer operator, finding the Fuss–Catalan algebra among the three [11].

Bulk integrability is half of what a model in a strip needs. Cherednik [12] and Sklyanin [13] identified the other half: a boundary operator must satisfy the reflection equation, and a solution of that equation together with a Yang–Baxter solution generates an infinite commuting family of double-row transfer matrices. For one colour this programme is finished. The general diagonal solutions of the boundary Yang–Baxter equation for Temperley–Lieb and dilute Temperley–Lieb models were obtained by Behrend and Pearce [14]; the algebra that governs a single strand touching a wall was identified as the blob algebra of Martin and Saleur [15], whose structure was studied in Ref. [16] and which was developed as the one-boundary Temperley–Lieb algebra in the XXZ and loop settings [17], with the two-boundary algebra following later [18]. The boundary operators themselves have been constructed in general form: Lima-Santos solved the boundary Yang–Baxter equations for the spin-s $\mathcal{U}_q[sl(2)]$ Temperley–Lieb model and found families carrying $2s(s + 1) + 3/2$ free parameters for half-integer s and $2s(s + 1) + 1$ for integer s [19], and

* yaozy@lzu.edu.cn

Avan, Kulish and Rollet constructed reflection matrices associated with Temperley–Lieb R-matrices [20], later extending the construction to R-matrices of Hadamard type [21]. The spectra of the resulting one-colour open chains and of their transfer matrices have been obtained by Bethe ansatz [22–24]. Not every integrable open chain is reached this way: Tong *et al.* construct boundary operators for a chain whose R-matrix has no crossing unitarity, where the reflection equation is unavailable [25].

Beyond one colour the boundary side had no algebraic formulation at all until Shigechi [26]. Realizing Fuss–Catalan algebras on generalized Dyck paths through noncrossing partitions, he defined the one- and two-boundary Fuss–Catalan algebras for general r, adjoining to the bulk generators a family of boundary generators $E^{(s)}$ that attach the innermost s strands of the boundary bundle to a wall, and he solved the associated reflection equation at two colours. What he found there is a branch carrying two boundary generators, whose coefficients are rational in the spectral parameter but involve a square root of a rational function of the loop weights. Above two colours the problem was open in every respect: a boundary ansatz at rank r carries r unknown functions, the bulk operator acquires one more coefficient with a pole of its own at each rank, and no relation between the boundary equations at consecutive ranks was known. Whether the extra generators are usable at all — whether a boundary can reach deep into the bundle — had not been decided even at three colours.

This paper settles every rank at once. On the two-site one-boundary algebra carrying Di Francesco's fundamental bulk operator, we determine every identity-normalized meromorphic boundary operator obeying the reflection equation and the inversion identity, for every $r \geq 3$ and for loop weights away from an explicit finite list of hypersurfaces. Three findings come out. First, the answer does not depend on the rank: the same four branches occur at every $r \geq 3$, and no new solution ever appears. Second, the support saturates at two colours — every coefficient from the third boundary generator upward vanishes identically, so a boundary operator can never reach past the second strand of the bundle. Third, the arithmetic of the answer is fixed: two of the four branches exist only over a quadratic extension of the field generated by the loop weights, and that extension is genuine at every rank. In particular the multi-colour boundary problem is far more rigid than the one-colour one, where the solutions come in families with many free parameters.

The proof has three parts. Rank three is done exhaustively, by expanding the reflection equation over a complete 36-element diagram basis and eliminating. An injective colour-tail homomorphism then shows that a rank-r problem whose top coefficient vanishes is not merely similar to, but exactly equivalent to, the rank-$(r-1)$ problem. What closes the induction is a no-go: at four colours an elimination forces one explicit rational function of the loop weights to vanish, which it does not, and a projection onto pure-boundary words carries every higher rank into that same contradiction.

Section II fixes the algebra, the bulk operator, and the reflection equation. Section III states the classification. Sections IV–VII prove it: the rank-three base case, the colour-lowering homomorphism, the top-colour obstruction, and the induction. Section VIII settles the inversion identity, the normalization at $w = 1$, and the minimal coefficient fields. Section IX describes the independent checks, and Section X gives the scope and the outlook.

II. THE ALGEBRA, THE BULK OPERATOR, AND THE REFLECTION EQUATION

A. Generators and loop weights

For any $r \geq 2$ let 1-$\mathbb{BFC}_2^{(r)}$ denote the one-boundary Fuss–Catalan algebra on two sites of Ref. [26], generated by bulk elements $E_1^{(s)}$ and boundary elements $E_2^{(s)}$ with $1 \leq s \leq r$. The bulk generator $E_1^{(s)}$ reconnects the innermost s coloured strands of the two site bundles; the boundary generator $E_2^{(s)}$ attaches the innermost s strands of the second bundle to the wall, as drawn in Fig. 1. The theorem below says that the third such generator and every one beyond it never appears with a nonzero coefficient.

Three scalars weight the closed loops that stacking creates: the bulk weight t, and the weights e and o carried respectively by even-to-odd and odd-to-even boundary loops, written τ, τ_e, τ_o in Ref. [26]. We parametrize them by two algebraically independent variables q and q_2,

$$t = -(q + q^{-1}), \qquad e = -(q_2 + q_2^{-1}), \qquad o = \frac{q}{q_2} + \frac{q_2}{q}, \tag{1}$$

which is the general solution of the single constraint the three weights of a Fuss–Catalan algebra must satisfy,

$$t^2 + e^2 + o^2 - teo - 4 = 0. \tag{2}$$

Two of the three loop weights may thus be chosen freely and the third is then determined up to the exchange encoded in (1). All coefficients below live in the field

$$\mathbb{F} = \mathbb{C}(q, q_2), \tag{3}$$

the smallest field containing the loop weights, localized at the denominators displayed as they arise. Four combinations of the weights recur often enough to be abbreviated once and for all,

$$T = t^2 - 1, \qquad L = t^2 - 2, \qquad A = te - o, \qquad B = to - e, \tag{4}$$

each of which is a polynomial in the loop weights and none of which vanishes identically. Products of two boundary generators at the same site reduce to a single one,

$$E_2^{(s)} E_2^{(u)} = e^{\min(s,u)} E_2^{(\max(s,u))}, \tag{5}$$

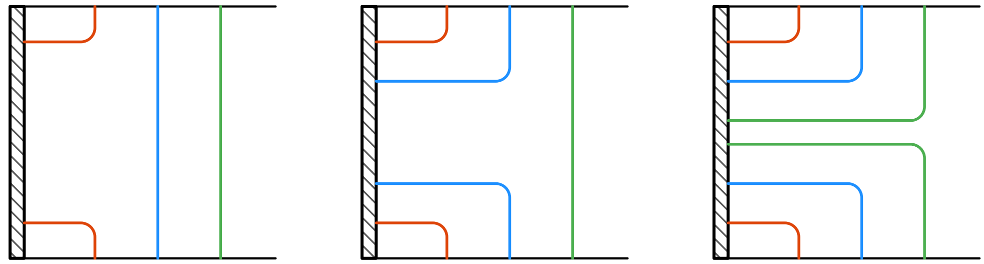

FIG. 1. The first three boundary generators of the Fuss–Catalan algebra, written $E(s)$ for $E_2^{(s)}$: each attaches the innermost one, two, or three coloured strands of the boundary bundle to the wall (hatched). These are the objects whose spectral-parameter-dependent coefficients are classified here. Theorem 1 states that $E(3)$, and every deeper generator at higher rank, carries the coefficient zero.

because stacking a shallower attachment onto a deeper one closes $\min(s,u)$ loops and leaves the deeper attachment behind; the same rule with e replaced by t holds at the bulk site.

One consequence of (5) is used repeatedly. For a boundary operator $1+\sum_s k_s(w)E_2^{(s)}$ define the cumulative quantities

$$\lambda_j(w) = 1+\sum_{s=1}^{j} e^s k_s(w), \qquad \lambda_0 = 1, \tag{6}$$

which are the eigenvalues the operator would have on a state where the innermost j strands are already attached to the wall: only the generators up to depth j act, and each contributes its coefficient times the loop weight it collects.

B. The bulk operator

A row-to-row transfer matrix collects the local Boltzmann weights along one row of the lattice, and the model is integrable when transfer matrices at different values of the spectral parameter commute, since the family then generates infinitely many conserved quantities. The standard sufficient condition is a one-parameter family of local operators $R_i(x)$ acting on the neighbouring sites $i, i+1$ and satisfying the Yang–Baxter equation, written here in the multiplicative convention of Refs. [9, 26],

$$R_i(x)\,R_{i+1}(xy)\,R_i(y) = R_{i+1}(y)\,R_i(xy)\,R_{i+1}(x). \tag{7}$$

Both sides move three world lines past one another, and the equation says that the order in which the crossings are performed does not matter.

The bulk structure used here is Di Francesco's fundamental branch [9]. Translating his generators into the normalization of Ref. [26] cancels an auxiliary square root and gives, for every $r \ge 2$,

$$R_i^{(r)}(x) = 1+\sum_{s=1}^{r-1}\frac{x^{s-1}(x-1)}{t^s}E_i^{(s)} + \frac{x^{r-1}(x-1)}{t^{r-2}(T-x)}E_i^{(r)}. \tag{8}$$

Every colour below the top contributes a coefficient that is a polynomial in the spectral parameter, and only the top colour carries a pole, at $x=T$; this single pole is what makes each new rank a genuinely new problem rather than a copy of the previous one. The operator satisfies $R_i^{(r)}(1)=1$ and $R_i^{(r)}(x)R_i^{(r)}(x^{-1})=1$, so it reduces to the identity at the symmetric point and undoes itself under inversion of the spectral parameter. Di Francesco's excited branches, which first exist at $r=4$, have different coefficients and define different boundary problems; they are outside the scope of this paper.

C. The reflection equation and the problem

A boundary operator acts at the wall and depends on its own spectral parameter. It preserves integrability when it satisfies the right reflection equation, Eq. (9.2) of Ref. [26],

$$\begin{aligned} &K_2^{(r)}(w)\,R_1^{(r)}\big((wz)^{-1}\big)\,K_2^{(r)}(z)\,R_1^{(r)}(w/z)\\ &\qquad = R_1^{(r)}(w/z)\,K_2^{(r)}(z)\,R_1^{(r)}\big((wz)^{-1}\big)\,K_2^{(r)}(w). \end{aligned} \tag{9}$$

Two world lines carrying spectral parameters w and z approach the wall, cross each other twice and bounce off the wall once each; the equation says that the two possible orders of those four events produce the same operator. By Sklyanin's construction [13], a solution of (9) together with (7) guarantees an infinite commuting family of double-row transfer matrices on a strip, one row of bulk operators carrying the system towards the wall and a second carrying it back.

The object to be classified is

$$K_2^{(r)}(w) = 1 + \sum_{s=1}^{r} k_s(w) E_2^{(s)}, \tag{10}$$

the general element of the boundary subalgebra with identity coefficient one, its r coefficients being meromorphic functions of w. Fixing the identity coefficient fixes the overall scale, which the reflection equation leaves free. We require, besides (9), that

$$K_2^{(r)}(1) = 1, \qquad K_2^{(r)}(w) K_2^{(r)}(w^{-1}) = 1, \tag{11}$$

so that the boundary is transparent at the symmetric point and undoes itself under inversion, matching the two properties the bulk operator already has. Classifying means more than exhibiting solutions: both sides of (9) must be reduced to a basis of the algebra and the resulting equations eliminated, so that the branches found are shown to exhaust the ansatz.

III. THE CLASSIFICATION

Theorem 1. *Let t, e, o be as in (1) with q, q_2 algebraically independent, let $\mathbb{F} = \mathbb{C}(q, q_2)$, fix an integer $r \geq 3$, and take the fundamental bulk operator (8). Then every $K_2^{(r)}$ of the form (10) with meromorphic coefficients satisfying the reflection equation (9) and the conditions (11), over $\mathbb{F}$ localized away from the zero loci listed in Sec. X, has*

$$k_s \equiv 0 \qquad \text{for every } s \geq 3, \tag{12}$$

so no boundary generator deeper than the second occurs, and is exactly one of the following.

(a) The identity. *$k_s \equiv 0$ for every s, that is $K_2^{(r)}(w) = 1$.*

(b) A one-constant family. *For a constant C independent of w,*

$$k_1(w) = \frac{L\,(w^2-1)}{\Xi_C(w)}, \qquad k_s(w) = 0 \quad (2 \leq s \leq r), \tag{13}$$

where $\Xi_C(w) = B + Cw - (eT - to)w^2$. This satisfies $K_2^{(r)}(1) = 1$ if and only if

$$C \neq t^2 e - 2to. \tag{14}$$

Set

$$\mathcal{D} = \frac{eoT\,A}{B}, \qquad S = \frac{aB}{T}, \qquad Q_a(w) = S - e\left\{aw^2 + (te - 2o)\,w\right\} \tag{15}$$

for either root a of $a^2 = \mathcal{D}$. Over the extension $\mathbb{F}(a)$ the families (a) *and* (b) *persist with their constants in the larger field, and exactly two further branches appear, one for each sign of a:*

(c,d) A conjugate pair.

$$k_1(w) = \frac{(aw - o)(w^2-1)}{w\,Q_a(w)}, \qquad k_2(w) = \frac{t\,(w^2-1)}{w\,Q_a(w)}, \tag{16}$$

with $k_s \equiv 0$ for $3 \leq s \leq r$. Each satisfies $K_2^{(r)}(1) = 1$ if and only if $Q_a(1) = S - e(a + te - 2o) \neq 0$, which holds for both signs on the generic locus.

The extension is genuine: $\mathcal{D}$ is not a square in $\mathbb{F}$. The minimal field over which each branch is defined is $\mathbb{F}$ for (a), *$\mathbb{F}(C)$ for a branch* (b) *with $C \neq 0$, and $\mathbb{F}(\sqrt{\mathcal{D}})$ for each of* (c,d).

Figure 2 summarizes the statement. Three features of it are worth separating. The list does not depend on r: the same four branches occur at three colours and at three hundred, and nothing new is gained by making the bundle deeper. The two conditions in (11) play different roles: $K_2^{(r)}(1) = 1$ selects which members of each family survive, while the inversion identity turns out to hold automatically on every branch the reflection equation produces, as Sec. VIII shows. And a member excluded by the normalization is not thereby singular: it can remain meromorphic and regular at $w = 1$ while taking a value there other than the identity.

Example 2. At $(q, q_2) = (2, 3)$ the loop weights (1) are $(t, e, o) = (-5/2, -10/3, 13/6)$. Taking $C = 7$ in (13) gives

$$k_1(w) = \frac{51\,(w^2-1)}{145w^2 + 84w - 25}, \qquad k_s = 0 \quad (2 \leq s \leq r),$$

whose denominator does not vanish at $w = 1$, so this member is normalized and regular. The proofs below are exact rather than numerical; the example is here only to make the statement concrete.

IV. THE RANK-THREE BASE CASE

A. A complete diagram basis

Reduction inside the algebra is governed, colour by colour, by three rules. For a single colour, writing e_1 for the bulk generator and b for the boundary generator,

$$e_1^2 = t\,e_1, \qquad b^2 = e\,b, \qquad e_1 b e_1 = o\,e_1, \tag{17}$$

each of which removes one letter in exchange for a loop weight: the first closes a bulk loop, the second a boundary loop, and the third the boundary loop that a strand makes by leaving the wall and returning.

Lemma 3. *The rewriting system (17) leaves exactly six one-colour normal words,*

$$1, \quad e_1, \quad b, \quad e_1 b, \quad b e_1, \quad b e_1 b, \tag{18}$$

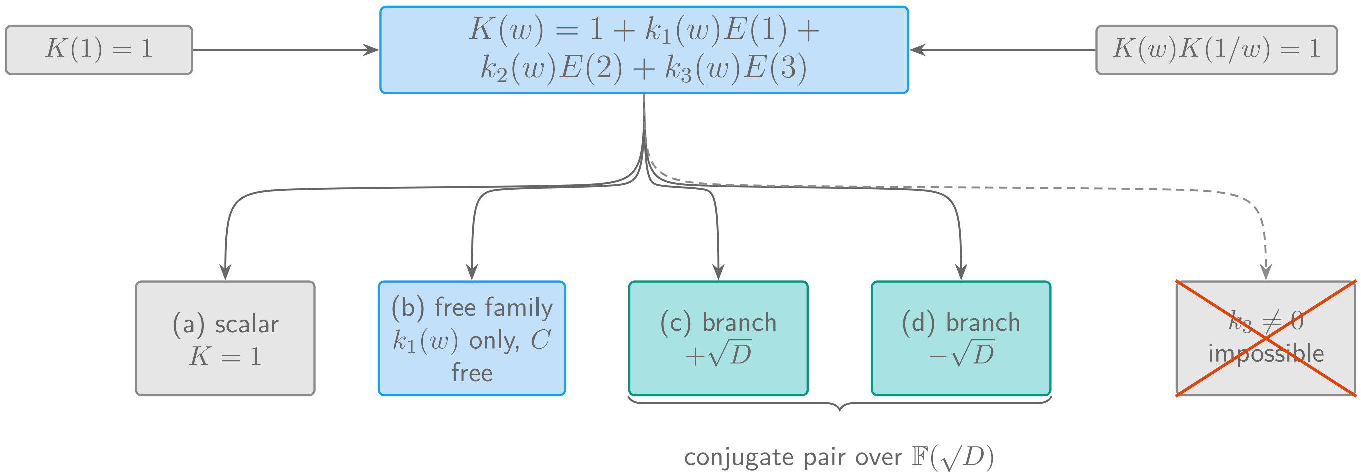

FIG. 2. The classification of Theorem 1, with $E(s)$ written for $E_2^{(s)}$ and D for $\mathcal{D}$. Whatever the rank $r \geq 3$, the identity-normalized solutions of the reflection equation are the identity, a family carrying one free constant C and supported on $E(1)$ alone, and a conjugate pair supported on $E(1)$ and $E(2)$ that exists only over the quadratic extension $\mathbb{F}(\sqrt{\mathcal{D}})$. The branch reaching the third generator, and every deeper one, is empty.

and the admissible three-colour triples of these number exactly 36*, which is the dimension of* 1-$\mathbb{BFC}_2^{(3)}$*. Hence the* 36 *normal triples are a basis, and two elements of* 1-$\mathbb{BFC}_2^{(3)}$ *are equal if and only if all* 36 *of their coefficients agree.*

Proof. A word in e_1 and b is normal when it contains no factor e_1e_1, bb, or e_1be_1; since the letters alternate in a normal word and the word cannot contain two consecutive equal letters, a normal word is determined by its length and its first letter, and the rules forbid length beyond three. That leaves the six words in (18). Nesting of the strand bundles restricts which triples occur: a letter of colour s may be followed by a letter of colour $s+1$ only when the shorter word ends on the boundary, and preceded by one only when the longer word begins there. Imposing this condition on the $6^3 = 216$ unrestricted triples leaves 36, as counted in Appendix A. Since every word reduces to one of the 36 and their number equals the dimension the algebra has by Proposition 7.3 of Ref. [26], no relation among them survives, and they form a basis. □

The point of Lemma 3 is that the count is exact rather than an upper bound. Equating 36 coefficients is therefore equality in the abstract algebra and not merely in some representation of it, a distinction that decides whether a classification is complete: a proof carried out inside a representation would leave open whether further solutions live in its kernel.

B. The coefficient inventory

Lemma 4. *Expanded in the basis of Lemma 3, the difference of the two sides of the reflection equation* (9) *at* $r = 3$ *has exactly* 20 *coefficients that are not identically zero. They occur in ten pairs related by exchanging* w *and* z *and reversing the overall sign, so the reflection equation is equivalent to the vanishing of ten functional equations in* k_1, k_2, k_3.

Proof. The proof is an exhaustive computation, carried out symbolically over $\mathbb{F}$ and over the field of rational functions in w and z.

Each case is one of the 36 normal triples of Lemma 3 — an explicit word such as `beb`|`eb`|`b`, one one-colour normal word for each of the three colours. The reflection difference is a single element of 1-$\mathbb{BFC}_2^{(3)}$ whose coefficients are rational functions of t, e, o, w, z and of the six unknowns $k_s(w), k_s(z)$; expanding it means applying the rules (17) and (5), together with the concatenation rule for mixed colours, until every product is a normal triple.

The test applied to a case is whether its coefficient reduces to the zero rational function. This is decisive because the rewriting system is confluent — the three rules shorten every word and overlap only in ways that reduce to the same normal word — so the reduced coefficient of a normal triple is unique, and by Lemma 3 the reflection difference vanishes in the algebra precisely when all 36 reduce to zero. The cases are exhaustive because Lemma 3 enumerates the basis in full: there are 36 normal triples and no others, and every product occurring in (9) reduces to a combination of them.

Sixteen coefficients reduce to zero identically. The remaining 20 do not, and inspection of them shows that each is carried to the negative of another by the substitution $w \leftrightarrow z$, which pairs them into ten classes. Since a function and its reversal vanish together, ten independent equations remain. □

Everything that follows at rank three is elimination on those ten equations. They split into three mutually exclusive and jointly exhaustive cases according to which of the coefficients vanish identically: $k_3 \not\equiv 0$; $k_3 \equiv 0$ with $k_2 \not\equiv 0$; and $k_2 \equiv k_3 \equiv 0$. We take them in turn.

C. No solution reaches the third colour

Proposition 5. *At $r = 3$ and on the generic locus, no solution of (9) has $k_3 \not\equiv 0$.*

Proof. Write $h = k_3$ and divide by it, which is legitimate because the branch $h \equiv 0$ has already been separated off and because an identity between meromorphic functions valid off the isolated zeros of h extends across them. The two coefficients of Lemma 4 carrying the shortest normal words give

$$k_2(u) = \frac{\gamma u - o}{t}\, h(u), \qquad k_1(u) = \left(\delta u^2 - \frac{o\gamma}{t^2}\, u\right) h(u) \tag{19}$$

for constants γ, δ that do not depend on the spectral parameter: each of those two coefficients states that a function of u takes the same value at w and at z, which forces it to be constant. So the two lower coefficients are determined by the top one up to two numbers.

Introduce

$$\begin{aligned} g(u) &= \frac{u^2 - 1}{u\, h(u)}, \\ \Psi(u) &= \delta t^2 (u^3 - u) + \gamma\, A\, u^2 + (t^2 e^2 - 2teo)\, u, \end{aligned} \tag{20}$$

the first being the natural variable in which the remaining equations linearize, and the second a cubic polynomial in the spectral parameter built from the two constants. A fifth coefficient becomes the separated statement $g(w) - g(z) = -(e/t^2)\{\Psi(w) - \Psi(z)\}$, whose complete meromorphic solution is

$$g(u) = \ell - \frac{e}{t^2}\Psi(u) \tag{21}$$

for a single constant ℓ; a difference depending only on u on each side fixes the function up to an additive constant. Substituting (21) into the successive coefficients then forces

$$\ell = 0, \qquad \gamma^2 = te\, A, \qquad \delta t^2 A = e^2 T(te - 2o), \tag{22}$$

so all three constants are pinned, the last two only up to the sign of γ.

These necessary conditions are inconsistent with a further coefficient. After imposing (22) its residual is a nonzero multiple of

$$H_{\mathrm{obs}} = \frac{H_1 H_2}{q^6 q_2^4}, \tag{23}$$

where H_1 and H_2 are the polynomials in $\mathbb{Z}[q, q_2]$ recorded in Appendix B, so the branch survives only where H_{obs} vanishes. Neither factor is the zero polynomial, which a single specialization certifies:

$$H_{\mathrm{obs}}(2, 3) = \frac{13438225}{5184} \neq 0. \tag{24}$$

A polynomial with one nonzero value is not the zero polynomial, and since q and q_2 are algebraically independent, $H_{\mathrm{obs}} \neq 0$ in $\mathbb{F}$. The branch is therefore empty. □

Proposition 5 is the step with no counterpart at one or two colours, where there is no third generator to use. It is what forbids a boundary operator from reaching all three strands of the bundle, and the reason the classification below is so much more rigid than its one-colour analogue.

D. The conjugate pair

Proposition 6. *At $r = 3$, on the generic locus and with $k_3 \equiv 0$, every solution with $k_2 \not\equiv 0$ is one of the two branches (16).*

Proof. One coefficient of Lemma 4 forces

$$k_1(u) = \frac{au - o}{t}\, k_2(u) \tag{25}$$

for a constant a, by the same argument that gave (19): the shallower coefficient is a fixed spectral-parameter-dependent multiple of the deeper one. Writing $g_2(u) = (u^2 - 1)/\{u\, k_2(u)\}$, the next functional equation separates and has the complete meromorphic solution

$$g_2(u) = \ell_2 - \frac{e}{t}\left\{au^2 + (te - 2o)u\right\}, \tag{26}$$

a quadratic in the spectral parameter with one free additive constant. The remaining coefficients are together equivalent to the two conditions

$$\ell_2 = \frac{aB}{tT}, \qquad a^2 B = eoT\, A, \tag{27}$$

which fix the additive constant to S/t and force $a^2 = \mathcal{D}$ with $\mathcal{D}$ as in (15). Substituting back through (25) and (26) gives exactly (16), and with those coefficients all 20 coefficients of Lemma 4 vanish identically, so these are solutions and not merely candidates. The two roots $a = \pm\sqrt{\mathcal{D}}$ give the two conjugate branches. □

E. The one-constant family and the identity

Proposition 7. *At $r = 3$, on the generic locus and with $k_2 \equiv k_3 \equiv 0$, the solutions are exactly the family (13) and the identity.*

Proof. Exactly one of the ten equations survives, namely

$$\begin{aligned} &-Lw(z^2 - 1)k_1(w) + Lz(w^2 - 1)k_1(z) \\ &\qquad + (w - z)\left\{(eT - to)wz + B\right\} k_1(w) k_1(z) = 0. \end{aligned} \tag{28}$$

It is a single quadratic functional equation relating the one remaining coefficient at two spectral parameters. Setting $g_1(u) = (u^2-1)/\{u\,k_1(u)\}$ turns it into the separated form

$$g_1(w) - g_1(z) = -\frac{w-z}{L}\left(eT - to + \frac{B}{wz}\right), \tag{29}$$

in which each side depends on w and z only through a difference of functions of one variable. Its complete meromorphic solution carries a single integration constant C and yields (13). If instead $k_1 \equiv 0$ the operator is the identity. The three branch splits are mutually exclusive and jointly exhaustive, and meromorphic continuation covers the isolated zeros used in each division, so together with Propositions 5 and 6 this classifies rank three. □

V. LOWERING THE RANK BY ONE COLOUR

Rank three is now settled, and the task is to reach every higher rank without expanding diagram bases whose dimension grows with r. The mechanism is a map that adds colours to the outside of the bundle, and it is exact rather than asymptotic.

Lemma 8. *Let $r = m + p$ with $m \geq 2$ and $p \geq 1$. Writing $E_i^{(j)}$ for the generators of $1\text{-}\mathbb{BFC}_2^{(m)}$ on the left and of $1\text{-}\mathbb{BFC}_2^{(r)}$ on the right, the assignment*

$$\Phi_p(E_1^{(j)}) = t^{-p}E_1^{(p+j)}, \qquad \Phi_p(E_2^{(j)}) = E_2^{(j)} \tag{30}$$

extends to an injective algebra homomorphism $1\text{-}\mathbb{BFC}_2^{(m)} \to 1\text{-}\mathbb{BFC}_2^{(r)}$. With

$$P_p(x) = 1 + \sum_{s=1}^{p} \frac{x^{s-1}(x-1)}{t^s} E_1^{(s)} \tag{31}$$

the fundamental bulk operators at the two ranks are related by

$$R^{(r)}(x) = P_p(x)\,\Phi_p\Big(R^{(m)}(x)\Big), \tag{32}$$

and $P_p(x)P_p(x^{-1}) = 1$. Consequently, for a boundary operator supported on the first m generators, the difference Δ_n of the two sides of the rank-n reflection equation obeys

$$\Delta_r(w,z) = P_p(v)P_p(u)\,\Phi_p(\Delta_m(w,z)) \tag{33}$$

with $u = w/z$ and $v = (wz)^{-1}$, so the rank-r reflection equation holds if and only if the rank-m one does, and the same for both conditions in (11).

Proof. Write a normal word of $1\text{-}\mathbb{BFC}_2^{(m)}$ in the tensor-word notation of Lemma 3. A word containing no bulk letter maps under Φ_p to $W \otimes 1^{\otimes p}$, while a word containing one maps to $t^{-p}W \otimes \mathsf{e}^{\otimes p}$; distinct normal words stay distinct, so Φ_p is injective. For multiplicativity, if both factors contain a bulk letter then the tail contributes $(\mathsf{e}^{\otimes p})^2 = t^p\,\mathsf{e}^{\otimes p}$, which cancels one of the two factors t^{-p}; the cases with one or no bulk-containing factor are immediate. So Φ_p embeds the smaller algebra as the sub-bundle of the p innermost colours.

For (32), the same-site product rule and the identity $1 + \sum_{s=1}^{p} t^s x^{s-1}(x-1)/t^s = x^p$ give, on the right-hand side, the coefficient $t^{-p}x^p\,x^{j-1}(x-1)/t^j$ for $E_1^{(p+j)}$ with $j < m$ and the top coefficient of (8) for $j = m$; together with the levels already in (31) these are every coefficient of the left-hand side. Thus the rank-r bulk operator splits into a part acting only on the p outer colours and the image of the rank-m operator.

The cumulative eigenvalues of $P_p(x)$, computed from (6) with e replaced by t, are x^j for $j \leq p$ and x^p beyond, all of which invert under $x \mapsto x^{-1}$; hence $P_p(x)^{-1} = P_p(x^{-1})$. Moreover P_p commutes with the embedded bulk operator and with every $E_2^{(j)}$ for $j \leq m$, since each relevant pair of depths satisfies $s + j \leq p + m = r$ and the two attachments then act on disjoint strands. Moving the commuting prefixes to the left in (9) gives (33). Since $P_p(v)P_p(u)$ is invertible and Φ_p injective, Δ_r vanishes exactly when Δ_m does, and because Φ_p fixes the boundary generators it carries $K^{(m)}(1) = 1$ and the inversion identity to their rank-r counterparts and back. □

The content of Lemma 8 is that the outer colours of the bundle are inert as long as the boundary does not reach them: a rank-r problem whose boundary stops at depth m is not merely similar to the rank-m problem, it is that problem. What remains is to show that the boundary never does reach the outer colours.

VI. NO BOUNDARY REACHES THE TOP COLOUR

A. The four-colour obstruction

Two bulk operators must be treated together. Besides the fundamental $R^{(4)}$ of (8), introduce the truncated four-colour operator

$$\widetilde{R}^{(4)}(x) = 1 + \sum_{s=1}^{4} \frac{x^{s-1}(x-1)}{t^s} E_1^{(s)}, \tag{34}$$

which differs from $R^{(4)}$ only in the top coefficient, having no pole. The truncated operator is not itself a bulk solution of interest; it is what the projection of the next subsection produces from every rank above four, so both must be excluded.

Proposition 9. *On the generic locus, neither the four-colour reflection problem for $R^{(4)}$ nor the one for $\widetilde{R}^{(4)}$ has an identity-normalized meromorphic solution with $k_4 \not\equiv 0$.*

TABLE I. The four coefficients of the four-colour reflection difference used in the proof of Proposition 9. Each row names the normal word whose coefficient is taken, the monomial in the spectral parameters whose coefficient inside it is extracted, and the factor that must then vanish. The factors are identical for the fundamental operator $R^{(4)}$ and for the truncated operator $\widetilde{R}^{(4)}$.

normal word	monomial	necessary factor
b|b|be|be	wz^2	$\ell_4 t^2$
b|b|be|e	w^3z^5	$e^2(\gamma_4^2 o - t^3 e - te^3 + 4te)$
b|be|be|be	w^2z^3	$t^2 e(\delta_4 A - te^3 + 2e^2 o)$
b|be|e|e	w^7z^6	$t^2 eo(\gamma_4 e - \delta_4 t)(\gamma_4 e + \delta_4 t)$

Proof. Set $h = k_4$ and $f_j = k_j/h$, dividing where the divisor is nonzero and extending meromorphically as before. For both bulk operators, the three coefficients of the reflection difference carrying the normal words beb|beb|beb|eb, beb|beb|eb|eb and beb|eb|eb|eb give successively

$$\begin{aligned} tzf_3(w) - twf_3(z) + o(z-w) &= 0, \\ t^2\{z^2 f_2(w) - w^2 f_2(z)\} + o\gamma_4 wz(z-w) &= 0, \\ t^3\{z^3 f_1(w) - w^3 f_1(z)\} + o\delta_4 t^2 w^2 z^2 (z-w) &= 0. \end{aligned} \tag{35}$$

Each equation says that one function of a single variable takes equal values at w and at z, hence is constant, so with constants $\alpha_4, \gamma_4, \delta_4$ independent of the spectral parameter,

$$\begin{gathered} f_3(x) = \frac{\gamma_4 x - o}{t}, \qquad f_2(x) = \delta_4 x^2 - \frac{o\gamma_4}{t^2}x, \\ f_1(x) = \alpha_4 x^3 - \frac{o\delta_4}{t}x^2. \end{gathered} \tag{36}$$

Each shallower coefficient is thus a fixed polynomial multiple of the top one, with one new constant appearing at each depth.

With $g_4(x) = (x^2-1)/\{x\,h(x)\}$ the coefficient of b|b|b|be separates and gives

$$g_4(x) = \ell_4 - \frac{e}{t^2}\Psi_4(x), \tag{37}$$

where

$$\begin{aligned} \Psi_4(x) = {}& \alpha_4 t^2(x^4 - x^2) + \gamma_4(te^2 - eo)x^2 \\ & + \delta_4(t^2 e - to)(x^3 - x) + (t^2 e^3 - 2te^2 o)x \end{aligned} \tag{38}$$

is a quartic polynomial in the spectral parameter assembled from the three constants. So the top coefficient is itself determined up to one further constant ℓ_4, and the entire boundary operator now depends on four numbers.

Four more coefficients suffice to contradict that. Substituting (36) and (37), clearing the bulk denominators displayed in (8) and (34) and reducing modulo the constraint (2), the coefficients listed in Table I carry the factors printed there, identically for the two bulk operators. On the generic locus, where t, e, o and A do not vanish, these force

$$\ell_4 = 0, \quad \gamma_4^2 = teA, \quad \delta_4 A = e^2(te - 2o), \quad \gamma_4^2 e^2 = \delta_4^2 t^2, \tag{39}$$

where the second uses $t^2 + e^2 - 4 = oA$, a rearrangement of (2). The three unknown constants are thereby overdetermined: eliminating γ_4 and δ_4 from the last three conditions leaves a relation among the loop weights alone,

$$H_4 = A^3 - te(te - 2o)^2 = 0. \tag{40}$$

That relation is false. In the parametrization (1),

$$\begin{aligned} H_4 = \frac{1}{q^3 q_2^3}\,(q^2 + q_2^2)\,\big(& q^4 q_2^4 + q^4 q_2^2 - q^4 \\ & + q^2 q_2^4 + q^2 - q_2^4 + q_2^2 + 1\big), \end{aligned} \tag{41}$$

and $H_4(2,3) = 21853/216 \neq 0$, so H_4 is not the zero element of $\mathbb{F}$. Hence no such solution exists for either bulk operator. □

The elimination in Proposition 9 rests on a finite exhaustive expansion of the same kind as Lemma 4, one rank higher. Each case is a four-colour normal word; the test is whether its coefficient reduces to zero; the reduction is decisive because the four-colour normal words are again a basis, obtained from the same rewriting system; and the cases are exhaustive because every product in the reflection equation reduces to a combination of them. The full expansion has 40 coefficients that are not identically zero, falling into 20 reversal pairs, for each of the two bulk operators; the four rows of Table I are the ones the argument uses, and the remaining coefficients are consistent with, and add nothing to, (39).

B. Transporting the obstruction to every rank

Lemma 10. *Let $r = p + 4$ with $p \geq 1$. Projecting the reflection difference onto normal words whose p outermost colours are exactly* b *and deleting that prefix gives*

$$\Pi_p(\Delta_r) = \frac{\lambda_p(w)\lambda_p(z)}{e^p}\,\Delta_{\widetilde{R}^{(4)}}\Big(\widehat{K}(w), \widehat{K}(z)\Big), \tag{42}$$

where λ_p is the cumulative quantity (6), the right side is the four-colour reflection difference for the truncated operator, and

$$\widehat{K}(x) = 1 + \sum_{j=1}^{4} \widehat{k}_j(x) E_2^{(j)}, \qquad \widehat{k}_j(x) = \frac{e^p k_{p+j}(x)}{\lambda_p(x)}. \tag{43}$$

Proof. No word containing the bulk letter `e` can reduce to a word of `b` alone: each rule `ee` $\mapsto$ $t\,$`e`, `bb` $\mapsto$ $e\,$`b` and `ebe` $\mapsto$ $o\,$`e` leaves a bulk letter behind whenever one is present. Every bulk level above the fourth therefore drops out of this sector, and the first four bulk levels of (8) become exactly (34) on the remaining colours: the projection removes the outer colours and, with them, the pole that distinguished the fundamental operator from the truncated one.

On the boundary side, a term of depth at most p multiplying the full prefix $\mathtt{b}^{\otimes p}$ contributes precisely the scalar λ_p by the same-site rule (5); two full-prefix terms contribute the factor e^p; and the contribution in which both boundary factors stay below the prefix is proportional to the commutator of the two bulk operators, which vanishes. Expanding the remaining cases gives (42). The identity normalization forces $\lambda_p(1) = 1$, so λ_p is not the zero function and the division in (43) is legitimate as an identity of meromorphic functions. $\square$

Proposition 11. *For every $r \geq 4$, every identity-normalized meromorphic solution of the rank-r reflection equation on the generic locus has $k_r \equiv 0$.*

Proof. At $r = 4$ this is Proposition 9. For $r \geq 5$, suppose $k_r \not\equiv 0$. Then $\widehat{k}_4 \not\equiv 0$ in (43), and Lemma 10 turns the solution into an identity-normalized solution of the four-colour problem for $\widetilde{R}^{(4)}$ with nonvanishing top coefficient, which Proposition 9 forbids. $\square$

VII. PROOF OF THE CLASSIFICATION

Proof of Theorem 1. Fix $r \geq 3$ and a solution. If $r \geq 4$, Proposition 11 gives $k_r \equiv 0$, and Lemma 8 with $m = r-1$ and $p = 1$ identifies the remaining boundary operator with a solution of the rank-$(r-1)$ problem, carrying both conditions (11) across. Iterating lowers the rank one colour at a time until $r = 3$ is reached, where Propositions 5, 6 and 7 leave exactly the branches (a)–(d), all of which have $k_3 \equiv 0$. Tracing the iteration back gives (12).

Conversely, each rank-three branch lifts. Applying Lemma 8 in the other direction with $m = 3$ and $p = r-3$ shows that the same coefficients k_1, k_2, with every deeper coefficient zero, solve the rank-r reflection equation, since Δ_r is the image of Δ_3 under an injective map composed with an invertible factor. So the four branches occur at every rank and nothing else does. The statements about normalization, the inversion identity and the minimal fields are proved in Sec. VIII. $\square$

VIII. INVERSION, NORMALIZATION, AND MINIMAL FIELDS

A. Inversion through cumulative eigenvalues

The inversion identity in (11) need not be imposed coefficient by coefficient.

Lemma 12. *Let U_j denote the coefficient of $E_2^{(j)}$ in $K_2^{(r)}(w)K_2^{(r)}(w^{-1}) - 1$. Then for every $1 \leq j \leq r$,*

$$e^j U_j(w) = \lambda_j(w)\lambda_j(w^{-1}) - \lambda_{j-1}(w)\lambda_{j-1}(w^{-1}), \quad (44)$$

so the inversion identity holds if and only if $\lambda_j(w)\lambda_j(w^{-1}) = 1$ for every j.

Proof. Multiply out using (5): the coefficient of $E_2^{(j)}$ collects every product $E_2^{(s)}E_2^{(u)}$ with $\max(s, u) = j$, which is exactly the difference of the two cumulative products displayed. Since $\lambda_0 = 1$, the right-hand side of the resulting system telescopes, and all U_j vanish precisely when each cumulative product is one. $\square$

The lemma reduces r conditions to a single statement about the eigenvalues of the boundary operator, and because the cumulative quantities stop changing once the deepest nonzero coefficient is passed, it makes the check independent of the rank.

B. The one-constant family

Only k_1 is nonzero on the family (13), so all the cumulative quantities coincide and

$$\lambda_1(w) = \cdots = \lambda_r(w) = \frac{w^2\,\Xi_C(w^{-1})}{\Xi_C(w)}, \quad (45)$$

an expression manifestly inverted by $w \mapsto w^{-1}$, since the numerator and denominator exchange up to the factor w^2. By Lemma 12 the inversion identity therefore holds for every value of C. Normalization at the symmetric point is equivalent to $\Xi_C(1) \neq 0$, that is to (14). At the excluded value the numerator and denominator of k_1 vanish together and the cancellation leaves the finite value $k_1(1) = -2/e$ rather than zero, so the coefficient is regular there and it is the normalization, not the regularity, that fails.

C. The conjugate pair

On a branch (16) the defining constraints (27) imply $aS = eo\,A$, and every residual of the inversion identity factors through that one combination:

$$\lambda_j(w)\lambda_j(w^{-1}) - 1 = \frac{e\,(w^2-1)^2\,\{aS - eo\,A\}}{w^2\,Q_a(w)\,Q_a(w^{-1})} = 0, \quad (46)$$

for $j = 1, 2$, and $\lambda_j = \lambda_2$ for every $j \geq 2$, so one computation covers every rank. Normalization is equivalent to $Q_a(1) \neq 0$. Where $Q_a(1) = 0$ the cancellation can leave finite nonzero coefficients, and on further exceptional subloci a pole, so the exclusion is not uniformly a regularity condition. Taking the two signs together,

$$Q_a(1)\,Q_{-a}(1) = -\frac{e\,H_{\mathrm{obs}}}{BT}, \quad (47)$$

which is the same obstruction (23) that emptied the third-colour branch. One polynomial therefore controls two apparently unrelated things: the absence of a branch reaching the third generator, and the existence of the two branches that do occur. Since $H_{\mathrm{obs}} \neq 0$ in $\mathbb{F}$, both signs are normalized on the generic locus. For the identity there is nothing to check.

D. Minimal coefficient fields

Written in the parameters (1), the discriminant (15) is

$$\mathcal{D} = \frac{(q_2^2+1)(q^2+q_2^2)(q^4+q^2+1)(q^2q_2^2+1)}{q^2q_2^2\,(q^4+q_2^2)}, \tag{48}$$

a ratio of products of distinct irreducible polynomials. The divisor $q_2 - i$ occurs in it with odd valuation, and a square has even valuation at every divisor, so $\mathcal{D}$ is not a square in $\mathbb{F}$ and the quadratic extension is genuine.

The fields named in Theorem 1 are minimal, not merely sufficient, because each branch determines its own constant rationally. On a family member (13) with $C \neq 0$,

$$C = \frac{L(w^2-1)/k_1(w) - B + (eT - to)w^2}{w}, \tag{49}$$

so any field containing the coefficient function contains C, while the branch is defined over $\mathbb{F}(C)$; the two inclusions force equality. On a conjugate branch,

$$a = \frac{t\,k_1(w)/k_2(w) + o}{w}, \tag{50}$$

so the coefficient field contains a, while (16) is defined over $\mathbb{F}(a)$. Hence the minimal fields are $\mathbb{F}$, $\mathbb{F}(C)$ and $\mathbb{F}(\sqrt{\mathcal{D}})$ as stated, and in particular the conjugate branches cannot be rewritten over $\mathbb{F}$ by any change of the free constant.

IX. INDEPENDENT VERIFICATION

The two computer-assisted steps, Lemma 4 and the expansion behind Proposition 9, were each carried out twice, by implementations written separately and in different computer algebra systems, and were compared only after both had finished. The second implementation recovered the same 20 support positions among the 36 normal triples, the same ten reversal pairs, and the same coefficient formulas and signs; at four colours it reconstructed the four factors of Table I independently.

A sharper check concerns the relation between the abstract algebra and the generalized-Dyck-path representation in which the algebra was originally realized. Introduce independent placeholders for the coefficients of the two bulk factors and the two boundary factors, so that the 20 coefficients of Lemma 4 become a vector $\boldsymbol{v}$ of polynomials in twelve placeholders, and let M be the 100×20 matrix collecting the vectorized 10×10 matrices of the 20 corresponding diagrams in the path action at three sites. The vector $\boldsymbol{y}$ of the 100 entries of the reflection difference in that representation is then $\boldsymbol{y} = M\boldsymbol{v}$ by construction. Exact elimination gives

$$\mathrm{rank}_{\mathbb{F}}\, M = 20, \tag{51}$$

and the pivot minor has determinant $-t^{14}e^{18}o^{17}A^{11}$, which is nonzero on the generic locus. Writing Θ for the selection of those 20 pivot rows and $G = (\Theta M)^{-1}\Theta$ for the resulting left inverse, one verifies $GM = I_{20}$ and hence $\boldsymbol{v} = G\boldsymbol{y}$, so the ideal generated by the 100 representation entries and the ideal generated by the 20 abstract coefficients coincide,

$$\langle y_1, \ldots, y_{100}\rangle = \langle v_1, \ldots, v_{20}\rangle. \tag{52}$$

The equations one would write down inside the path representation are therefore neither more nor fewer than the abstract ones, and the classification could have been derived in either place.

Equation (52) should not be over-read. It does not say the path action is faithful on all 36 diagrams; the full image has rank 30, and six exact kernel relations exist. What it says is that the action is injective on the 20-dimensional support of the reflection difference, which is all the argument needs, since the kernel meets that support trivially.

Finally, the structural identities were replayed at finite rank as falsifiers. The factorization (32) was checked at 21 pairs (r, m) through rank eight and the homomorphism property of (30) on complete normal bases; the projection (42) was checked with arbitrary boundary coefficients at ranks five and six; and the four branches were verified entry by entry, at two exact rational choices of the loop weights, for every rank from two to eight, together with deliberate perturbations of the top coefficient, which the checks detect. These finite tests can only refute; the statement for unbounded r rests on Lemma 8 and Proposition 11.

X. DISCUSSION

A. Why the support stabilizes

Two separate things make the answer independent of the rank. The obstruction H_4 of (40) excludes a boundary reaching the outermost colour, and it does so uniformly, because the projection of Lemma 10 sends every rank to the same four-colour problem; it is a nonzero rational function of the loop weights, not a degeneracy of the ansatz. Once the top coefficient is known to vanish, the factorization of Lemma 8 removes that colour by an injective algebra map, without approximation. Alternating the two steps walks any rank down to three, which is why no diagram basis of growing dimension ever has to be expanded.

Two features of the two-colour solution therefore survive the whole hierarchy. The surviving functional equations separate under the substitution $g(u) = (u^2 - 1)/\{u\, k(u)\}$, which is what makes the complete meromorphic solution accessible rather than merely a rational ansatz; and the nonsquare discriminant is unavoidable, so the algebraic branches always come as a conjugate pair rather than as one branch over the field of the loop weights. Against the one-colour case the contrast is sharp: there the reflection matrices form families with a number of free parameters growing with the spin [19], whereas here, at every rank, one free constant and one sign is all there is.

B. Scope and exceptional loci

The theorem classifies the identity-normalized ansatz (10) modulo the scalar gauge; it does not classify arbitrary elements of the boundary algebra. It holds away from the zero loci of t, e, o, T, L, A, B, H_{obs}, H_4, the bulk and branch denominators displayed above, and the determinant of the rank certificate of Sec. IX. Each of those hypersurfaces can in principle carry additional solutions and requires a separate treatment in which the relevant quantity is not inverted. Di Francesco's excited bulk branches, which begin at four colours, define different reflection equations and lie outside the theorem as well.

One convention differs from the source. We read the second bulk coefficient in Eq. (9.12) of Ref. [26] at the argument w/z where that reference prints wz; the reflection equation (9), as stated there, requires the former, and with the printed argument the two-colour solution of that reference does not satisfy it.

C. Conclusion and outlook

For every rank $r \geq 3$ and generic loop weights, the identity-normalized boundary operators compatible with Di Francesco's fundamental bulk branch are the identity, a one-constant family carrying the first boundary generator alone, and a conjugate pair carrying the first two, the pair existing only over a genuine quadratic extension of the field generated by the loop weights. No solution ever reaches the third generator, so the boundary of a Fuss–Catalan lattice model cannot see more than two strands of the bundle however many strands there are. The consequence for the models is immediate: the double-row transfer matrices one can build on these bulk weights carry one continuous boundary parameter and one discrete choice, and no more.

Three directions follow. The two-boundary hierarchy is untouched; at one colour the step from one wall to two demanded a representation theory of its own [18], and the higher-colour algebras are already available [26]. Di Francesco's excited bulk branches need their own boundary classifications, and the mechanism used here does not apply to them unchanged, since the colour-lowering factorization is a property of the fundamental coefficients. Finally, the double-row transfer matrices built from Theorem 1 have not been diagonalized; at one colour the corresponding spectra have been obtained by Bethe ansatz [22–24], and here the free constant C plays the role of a boundary field whose effect on the spectrum is the natural next question. The generalized-Dyck-path realization, used above only as a check, may be the right tool for it.

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OpenAI GPT-5.6 Sol was used substantively in this work: in solving the problem, including the formulation of arguments and the derivation of intermediate steps, and in drafting and revising the manuscript. The authors specified the problem, guided the approach to be taken, checked every argument and result it produced, and take full responsibility for the content of this manuscript.

CODE AVAILABILITY

Two steps of the proof are computer assisted, and only two: the expansion of the rank-three reflection difference over the 36-element basis in Lemma 4, and the four-colour expansion behind Proposition 9. Both are finite exhaustive symbolic computations whose objects, reduction rules, and covering arguments are specified in the text above, so the proof can be followed without running anything; the programs let a reader repeat the expansions faster than by hand. They are available in the repository accompanying this article, together with the second, independently written implementations, the finite replays reported in Sec. IX, and the rank certificate of (52). The computations used Python 3.12 with SymPy 1.14 and Sage 10.9. No experimental data were generated or analysed.

Appendix A: The 36 normal forms

The rules (17) leave the six one-colour normal words of (18), which may be indexed by two bits: whether the word begins with the boundary letter and whether it ends with it. A three-colour normal form is a triple of such words compatible with the nesting of the strand bundles, the compatibility condition being that a colour-s letter may be followed by a colour-$(s+1)$ letter only when the shorter word ends on the boundary, and preceded by one only when the longer word begins there. Imposing this odd-last and even-first condition on the $6^3 = 216$ unrestricted triples leaves exactly 36, matching the dimension that Proposition 7.3 of Ref. [26] assigns to 1-$\mathbb{BFC}_2^{(3)}$. Because the count is exact rather than an upper bound, equating all 36 coefficients of the reflection difference is equality in the abstract algebra rather than in a representation of it.

Appendix B: The obstruction polynomial

The factor (23) that empties the third-colour branch is $H_{\mathrm{obs}} = H_1 H_2/(q^6 q_2^4)$ with

$$H_1 = q^4 q_2^4 + q_2^4 + 2q^4 q_2^2 + 2q^2 q_2^2 + q^6 + q^2, \tag{B1}$$

$$\begin{aligned} H_2 = {} & q^6 q_2^4 + q^4 q_2^4 + q^2 q_2^4 - q_2^4 + q^6 q_2^2 \\ & + q^4 q_2^2 + q^2 q_2^2 + q_2^2 - q^6 + q^4 + q^2 + 1. \end{aligned} \tag{B2}$$

Both are nonzero elements of $\mathbb{Z}[q, q_2]$, which the single value in (24) certifies. The same product reappears in (47) as the two-sign norm of the normalization denominator on the conjugate branches, so one polynomial governs both the exclusion of the third-colour branch and the survival of the branches that do exist.

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